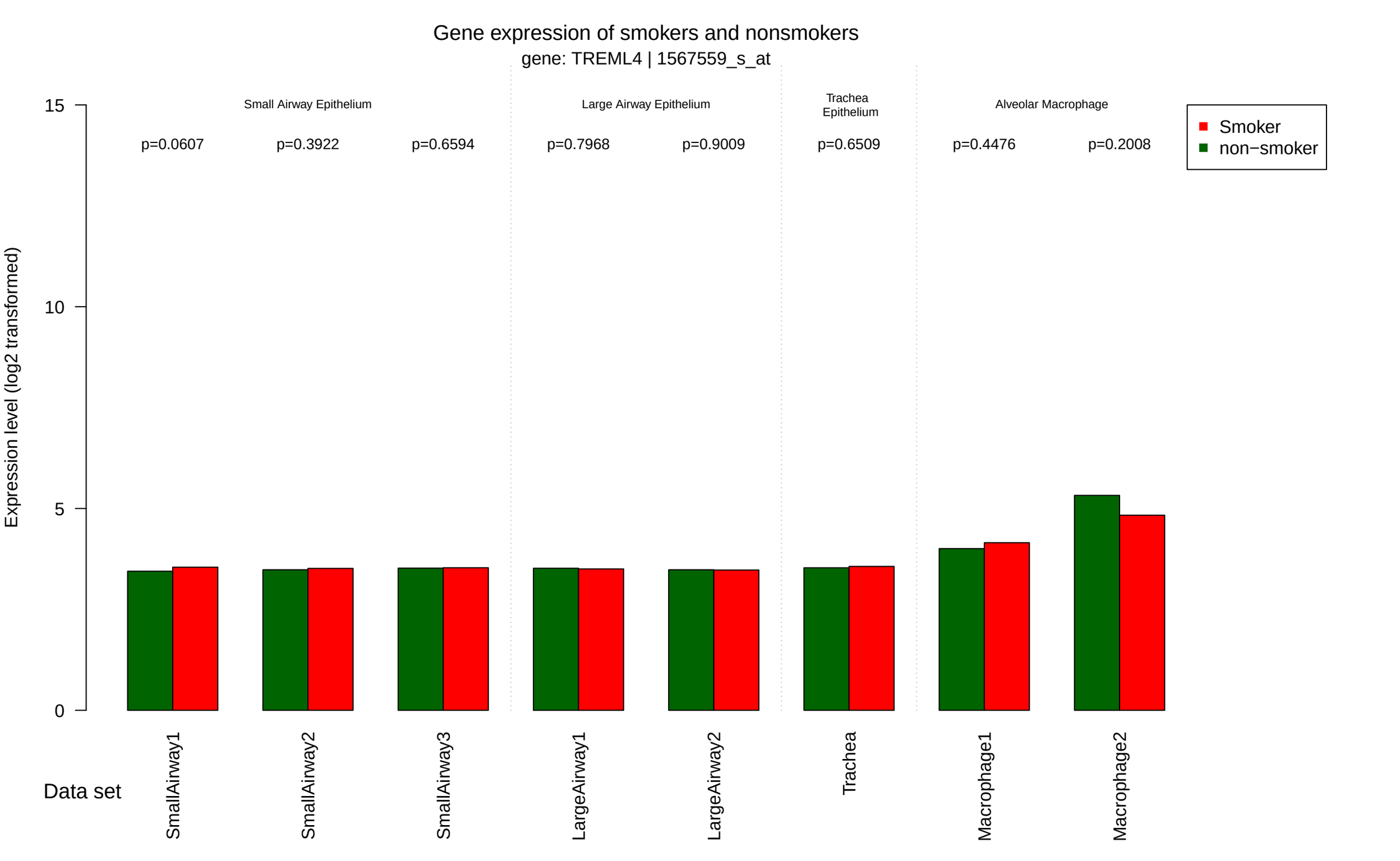
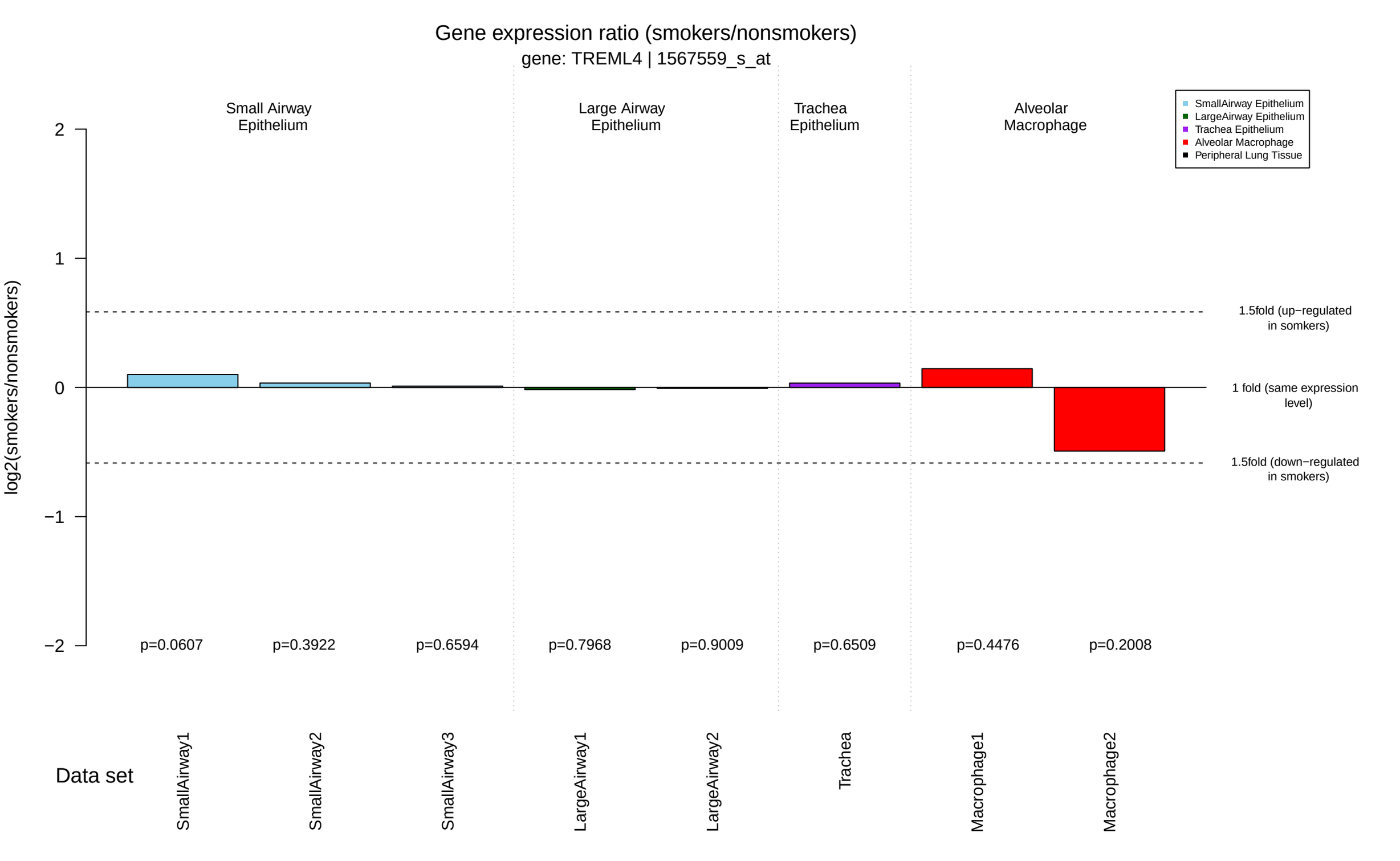
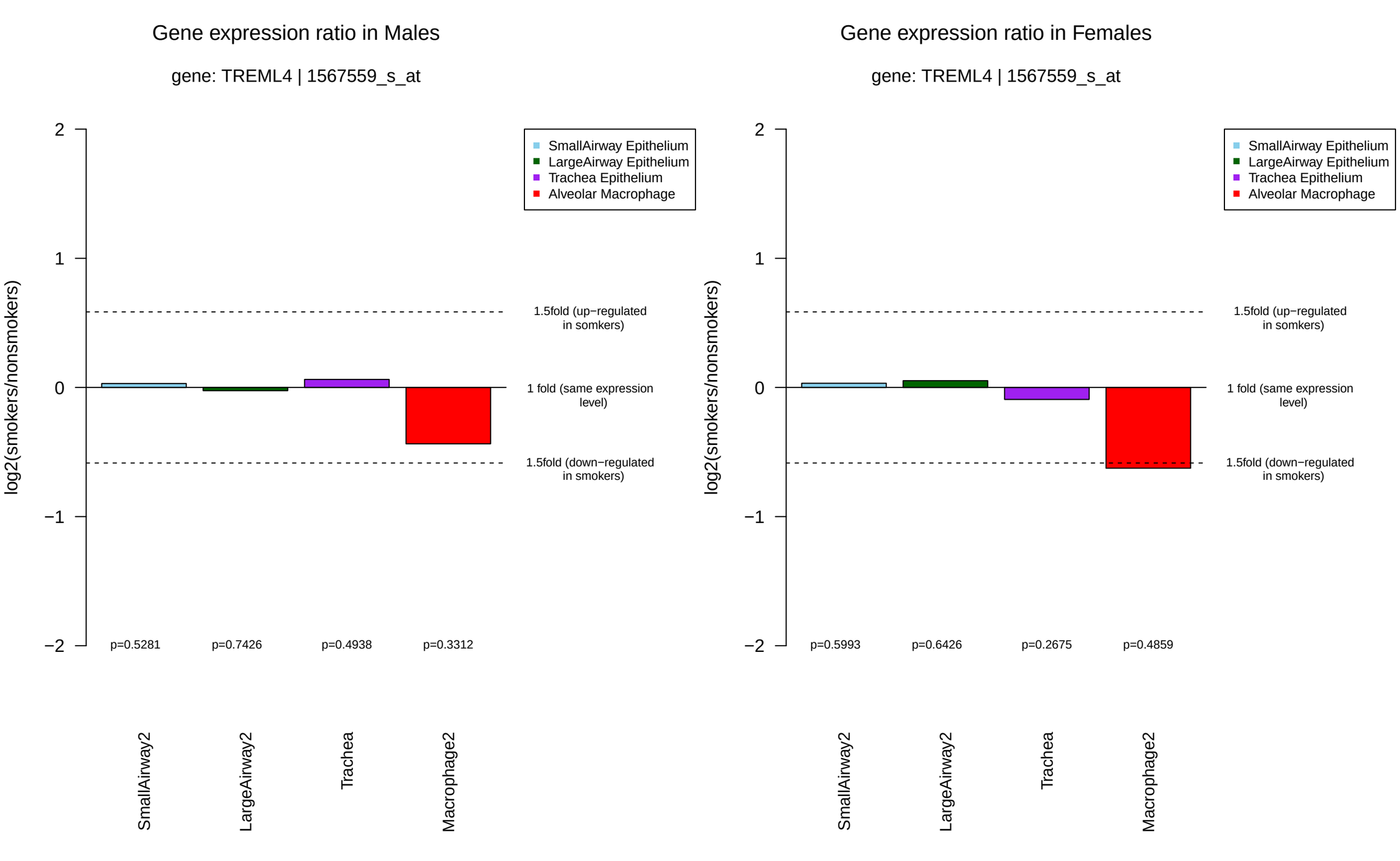
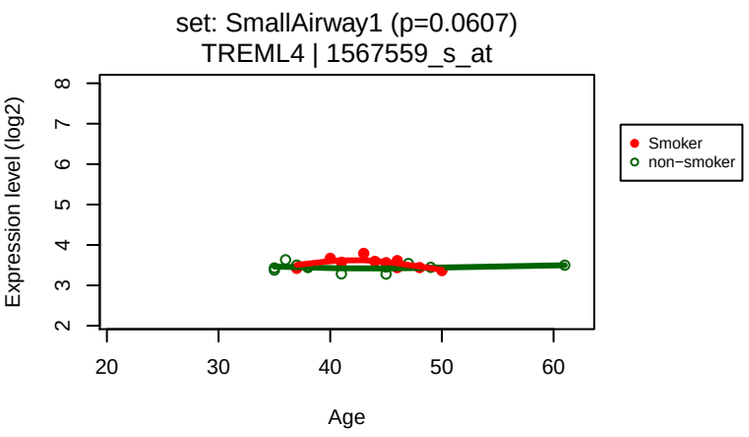
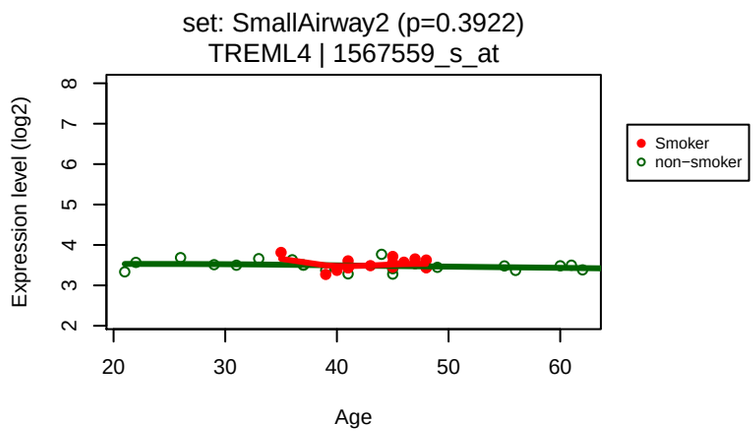
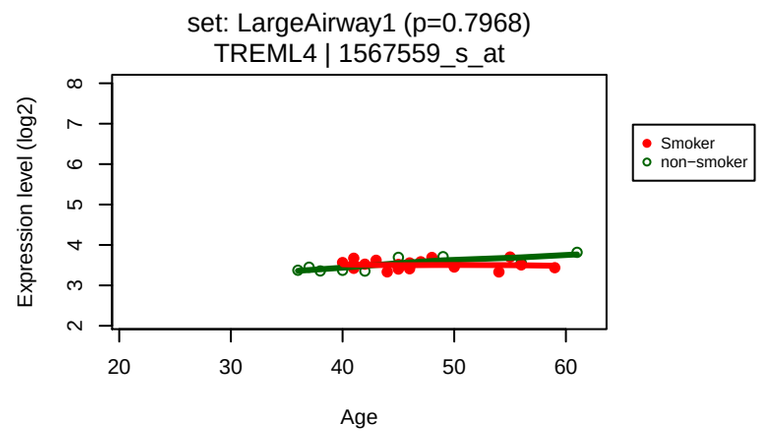

Gene expression of smokers and nonsmokers
gene: TREML4 | 1567559_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0607
p=0.3922
p=0.6594
p=0.7968
p=0.9009
p=0.6509
p=0.4476
p=0.2008
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

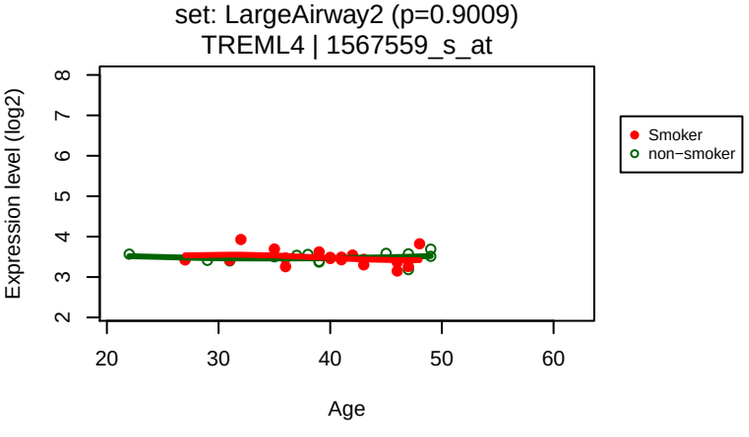

Gene expression ratio (smokers/nonsmokers)
gene: TREML4 | 1567559_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0607
p=0.3922
p=0.6594
p=0.7968
p=0.9009
p=0.6509
p=0.4476
p=0.2008
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

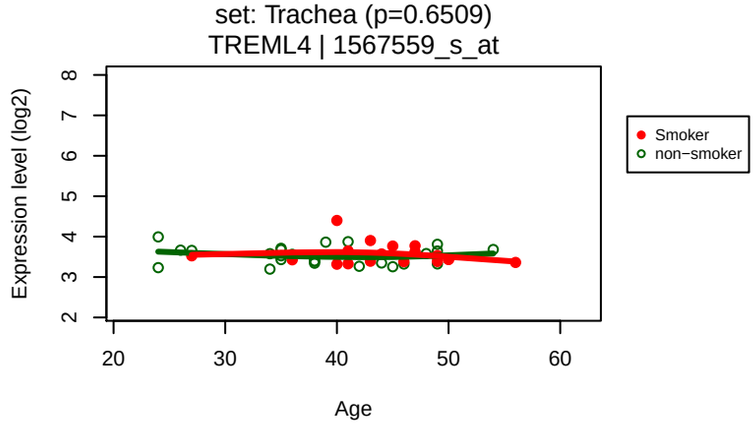

Gene expression ratio in Males
gene: TREML4 | 1567559_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5281
p=0.7426
p=0.4938
p=0.3312
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TREML4 | 1567559_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5993
p=0.6426
p=0.2675
p=0.4859
SmallAirway2
LargeAirway2
Trachea
Macrophage2

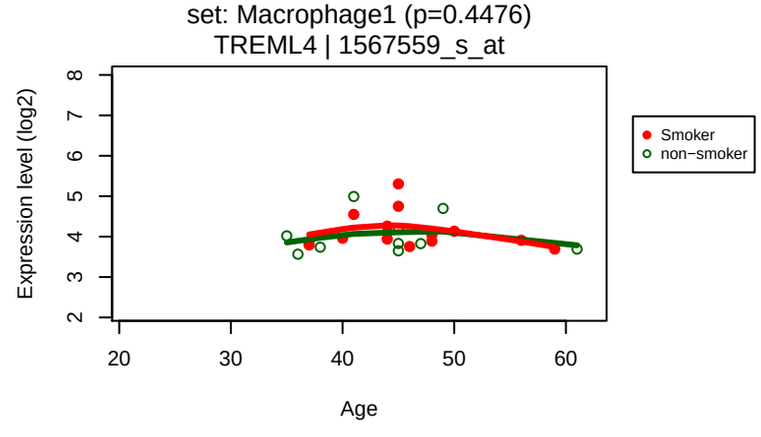

set: SmallAirway1 (p=0.0607)
TREML4 | 1567559_s_at
Expression level (log2)
Age
Smoker
non−smoker

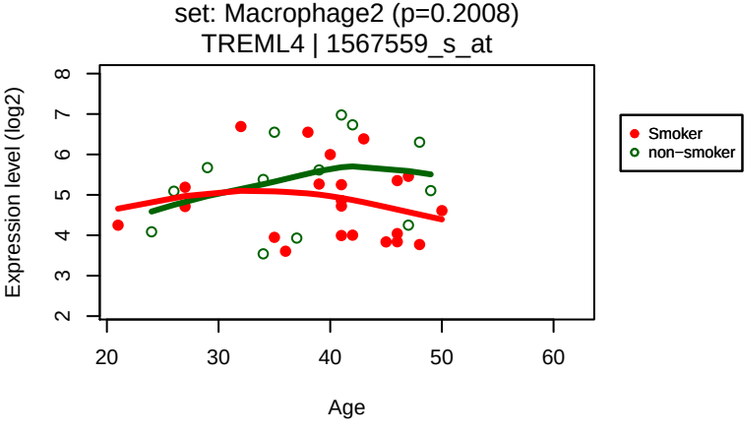

set: SmallAirway2 (p=0.3922)
TREML4 | 1567559_s_at
Expression level (log2)
Age
Smoker
non−smoker

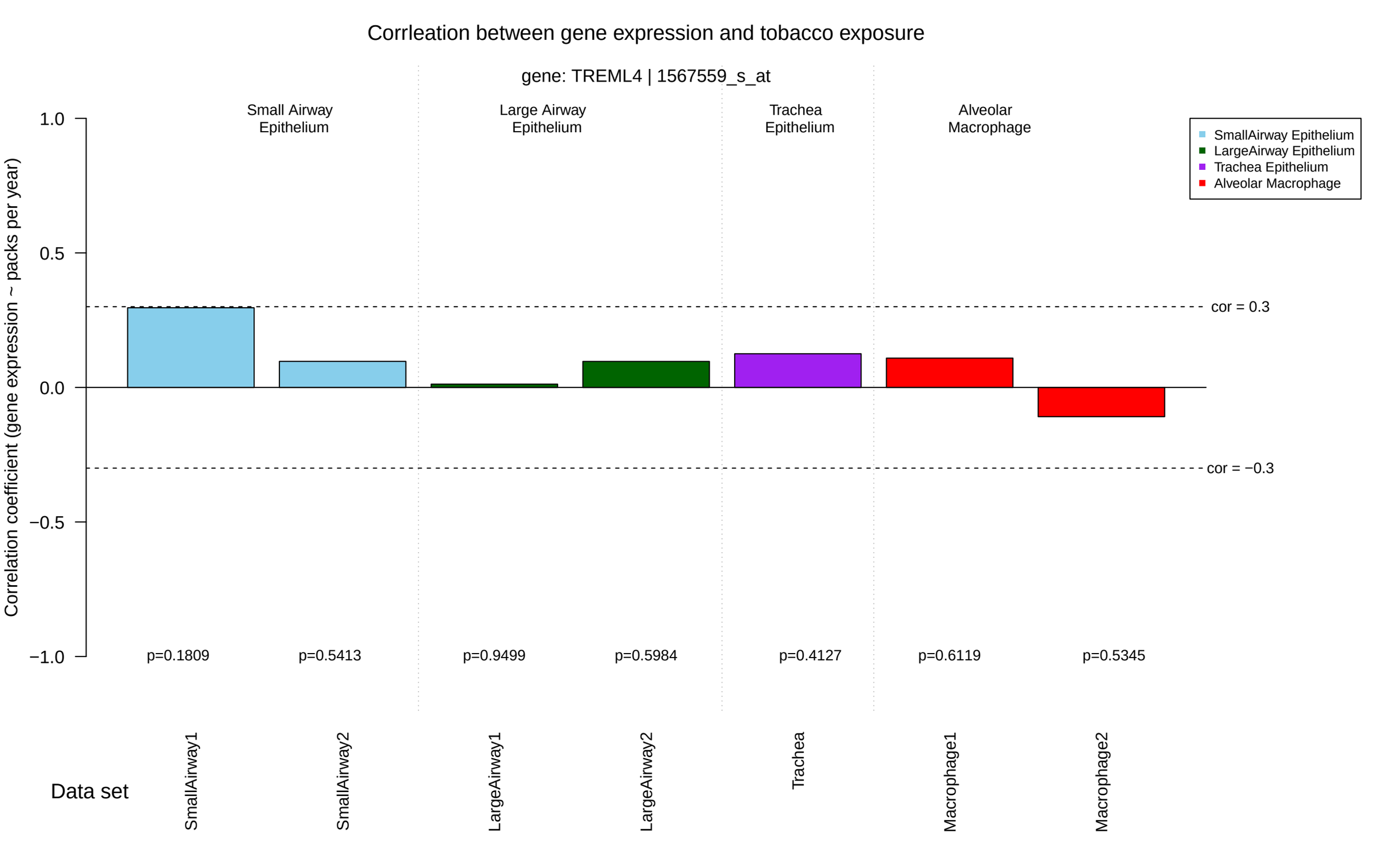

set: LargeAirway1 (p=0.7968)
TREML4 | 1567559_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9009)
TREML4 | 1567559_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6509)
TREML4 | 1567559_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4476)
TREML4 | 1567559_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2008)
TREML4 | 1567559_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TREML4 | 1567559_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1809
p=0.5413
p=0.9499
p=0.5984
p=0.4127
p=0.6119
p=0.5345
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2